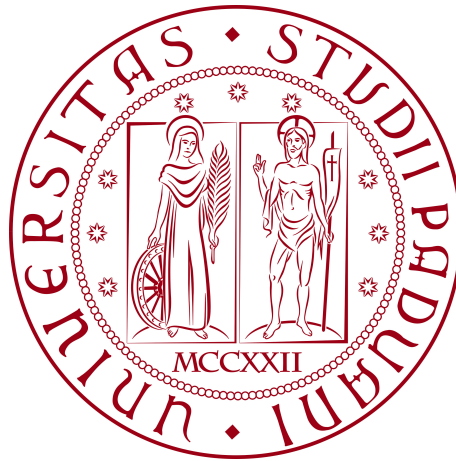


University of Padua

DEPARTMENT OF MATHEMATICS “TULLIO LEVI-CIVITA”

MASTER DEGREE IN COMPUTER SCIENCE



Master's Thesis

Supervisor

Prof. Luciano Serafini

Co-Supervisor

Prof. Roberto Confalonieri

Candidate

Marco Brugin

ID Number: 2122864

ACADEMIC YEAR 2026-2027

© Marco Brugin, 2027. All rights reserved. Master's Thesis: “”, University of Padua, Department of Mathematics “Tullio Levi-Civita”. Certo. Ti lascio una versione completa e coerente, con la citazione di Turing e un tono personale ma adatto a una tesi magistrale. Ho cercato di evitare frasi troppo enfatiche, mantenendo invece il focus sul percorso nel campo dell'AI e sulle persone che ti hanno sostenuto.

“We can only see a short distance ahead, but we can see plenty there that needs to be done.” — Alan Turing

Acknowledgements

First and foremost, I would like to express my sincere gratitude to Prof. Serafini and Prof. Confalonieri for their guidance and support throughout my Master’s degree. Their knowledge, advice, and availability have been fundamental to my academic growth and to the development of this thesis. I am particularly grateful for the opportunity to work in the field of Artificial Intelligence, a rapidly evolving area that has allowed me to deepen my technical knowledge while developing a greater understanding of the opportunities and challenges that arise from the development of intelligent systems.

I would also like to thank my entire family for their constant support throughout these years. Their presence, encouragement, and love have accompanied me throughout this journey, especially during the more difficult moments. Completing this degree required commitment, perseverance, and many sacrifices, and I am deeply grateful to my family for always being there for me and for supporting me in every step of the way.

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This thesis marks the conclusion of an important chapter of my academic journey and, at the same time, the culmination of a path that has allowed me to explore Artificial Intelligence in greater depth. During these years, I have had the opportunity to learn not only about algorithms, models, and technologies, but also about the broader questions surrounding the development and use of intelligent systems. The rapid evolution of Artificial Intelligence

makes it difficult to predict where this field will lead in the future, but it also makes clear that there is still much to discover, understand, and build.

As I reach the end of this Master's degree, I look back with gratitude at the people who supported me and at the experiences that shaped this journey. I hope to carry forward the knowledge, curiosity, and passion developed during these years, and to continue exploring the possibilities of a field that is still only beginning to reveal its full potential.

Padova, 2027

Marco Brugin

Abstract

This thesis investigates the use of Structured Logic Tensor Networks (LTNs) for neuro-symbolic learning. The goal is to combine neural learning with explicit background logical knowledge. The proposed approach addresses an AI classification task with limited and heterogeneous training data. Entities and concepts are represented through learnable continuous embeddings. Logical relations and domain constraints are expressed using first-order logic. These logical formulas are converted into differentiable constraints using fuzzy logic semantics. The model jointly learns from labeled examples and the available background knowledge. A structured representation is used to organize concepts, relations, and logical dependencies. This allows domain knowledge to guide the learning process rather than relying only on observed data. The approach is evaluated against conventional neural network baselines. Experiments investigate both predictive accuracy and logical consistency of the learned model. Additional experiments analyze the effect of increasing or reducing the amount of background knowledge. The results examine whether logical constraints improve generalization, particularly with limited training data. The thesis also discusses computational and modeling challenges associated with differentiable logical reasoning. Overall, the study demonstrates the potential of Structured LTNs for integrating learning and reasoning in AI systems.

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